Test Series No.

1.9

MASTERY TEST SERIES

DURATION: 180 Minutes | **M. MARKS: 720**

DATE: 10/03/2026

ANSWER KEY

1.	(1)	46.	(3)	91.	(3)	136.	(3)
2.	(2)	47.	(4)	92.	(2)	137.	(4)
3.	(4)	48.	(3)	93.	(1)	138.	(3)
4.	(2)	49.	(1)	94.	(4)	139.	(4)
5.	(1)	50.	(1)	95.	(1)	140.	(2)
6.	(2)	51.	(4)	96.	(3)	141.	(2)
7.	(3)	52.	(4)	97.	(2)	142.	(2)
8.	(4)	53.	(4)	98.	(4)	143.	(2)
9.	(2)	54.	(2)	99.	(3)	144.	(2)
10.	(2)	55.	(1)	100.	(1)	145.	(4)
11.	(1)	56.	(4)	101.	(3)	146.	(3)
12.	(3)	57.	(1)	102.	(2)	147.	(3)
13.	(1)	58.	(2)	103.	(4)	148.	(4)
14.	(4)	59.	(1)	104.	(1)	149.	(3)
15.	(3)	60.	(2)	105.	(2)	150.	(2)
16.	(2)	61.	(2)	106.	(3)	151.	(1)
17.	(4)	62.	(1)	107.	(4)	152.	(3)
18.	(2)	63.	(1)	108.	(1)	153.	(3)
19.	(2)	64.	(4)	109.	(4)	154.	(1)
20.	(3)	65.	(2)	110.	(3)	155.	(3)
21.	(2)	66.	(4)	111.	(2)	156.	(2)
22.	(2)	67.	(4)	112.	(1)	157.	(4)
23.	(3)	68.	(1)	113.	(3)	158.	(3)
24.	(1)	69.	(1)	114.	(4)	159.	(2)
25.	(2)	70.	(3)	115.	(4)	160.	(1)
26.	(1)	71.	(3)	116.	(3)	161.	(4)
27.	(2)	72.	(3)	117.	(2)	162.	(3)
28.	(2)	73.	(1)	118.	(1)	163.	(4)
29.	(1)	74.	(3)	119.	(4)	164.	(1)
30.	(3)	75.	(2)	120.	(2)	165.	(4)
31.	(3)	76.	(2)	121.	(4)	166.	(2)
32.	(2)	77.	(2)	122.	(1)	167.	(3)
33.	(2)	78.	(1)	123.	(3)	168.	(2)
34.	(2)	79.	(3)	124.	(4)	169.	(1)
35.	(3)	80.	(1)	125.	(1)	170.	(4)
36.	(3)	81.	(2)	126.	(2)	171.	(4)
37.	(3)	82.	(2)	127.	(3)	172.	(3)
38.	(2)	83.	(2)	128.	(2)	173.	(3)
39.	(2)	84.	(1)	129.	(4)	174.	(3)
40.	(2)	85.	(2)	130.	(1)	175.	(4)
41.	(1)	86.	(3)	131.	(3)	176.	(3)
42.	(2)	87.	(3)	132.	(4)	177.	(4)
43.	(1)	88.	(3)	133.	(2)	178.	(2)
44.	(1)	89.	(4)	134.	(4)	179.	(3)
45.	(2)	90.	(1)	135.	(1)	180.	(2)

[Contd...

Hints & Solutions

1. (1)

The C.O.M. can have coordinates (0, 1) and (0, –1)

$$Y_{COM} = \frac{M_1Y_1 + M_2Y_2}{M_1 + M_2}$$

$$1 = \frac{(2)(5) + 4(Y_2)}{6}$$

$6 = 10 + 4Y_2$

$-4 = 4Y_2$

$Y_2 = -1$

And $Y_{COM} = -1$

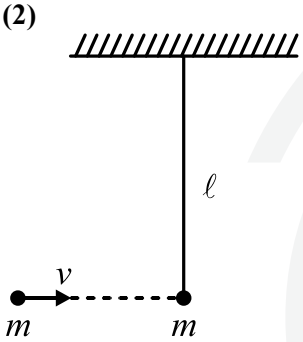
$$-1 = \frac{10 + 4Y_2}{6}$$

$$-16 = 4Y_2$$

$$Y_2 = -4$$

(NCERT Ed. 2025-26, 11th, Page No. 96)

2. (2)



Using conservation of linear momentum after collision speed of combined mass $= \frac{v}{2}$.

$$\frac{v}{2} = \sqrt{5g\ell}$$

$$v = 2\sqrt{5g\ell}$$

(NCERT Ed. 2025-26, 11th, Page No. 79)

3. (4)

$$H_1 = \frac{u^2 \sin^2\theta}{2g}$$

$$H_2 = \frac{u^2 \cos^2\theta}{2g}$$

$$H_2 - H_1 = \frac{u^2}{2g}(\cos^2\theta - \sin^2\theta)$$

$$\Rightarrow \cos^2\theta - \sin^2\theta = \frac{200}{400} = \frac{1}{2}$$

$$\Rightarrow \cos 2\theta = \frac{1}{2}$$

$$\Rightarrow 2\theta = \frac{\pi}{3}$$

$$\therefore \theta = \frac{\pi}{6} \text{ or } 30°$$

other angle is 60°

(NCERT Ed. 2025-26, 11th, Page No. 40)

4. (2)

$S_{rel} = u_{rel}t$

$(v - 10)(40) = 100$

$v - 10 = 2.5$

$\Rightarrow v = 12.5\,\text{m/s}$

(NCERT Ed. 2025-26, 11th, Page No. 17)

5. (1)

$x(\text{N}) = (1)u_2$

$\Rightarrow x\left[(\text{Kg})(\text{m})(s)^{-2}\right] = (1)\left[10^{-2}\text{Kg}\ 10^{-1}\text{m}\ 10^{2}s^{-2}\right]$

$\Rightarrow x = 0.1$ N

(NCERT Ed. 2025-26, 11th, Page No. 10)

6. (2)

$f_k = \mu N = 0.6 \times 20$

= 12 N

(NCERT Ed. 2025-26, 11th, Page No. 61)

7. (3)

$$T = 2\pi\sqrt{\frac{L\cos\theta}{g}}$$

$$T = 2\pi\sqrt{\frac{5 \times 0.8}{10}}$$

$$T = 2\pi\sqrt{\frac{4}{10}} = \frac{4\pi}{\sqrt{10}}$$

$T = 4$ second

(NCERT Ed. 2025-26, 11th, Page No. 63)

8. (4)

$$\frac{d^2x}{dt^2} + 8x = 0$$

On comparing with standard equation

$$\omega = \sqrt{8} = \frac{2\pi}{T}$$

$$T = \frac{2\pi}{2\sqrt{2}} = \frac{\pi}{\sqrt{2}}\,\text{s}$$

(NCERT Ed. 2025-26, 11th, Page No. 267)

9. (2)

$PV^2 = \text{Constant}$

$$\left(\frac{nRT}{V}\right)^2 \times V = \text{Constant}$$

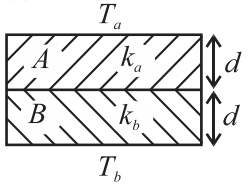
$$\frac{T^2}{V} = \text{Constant}$$

$\therefore T_1 < T_2 \Rightarrow V_1 < V_2$

(NCERT Ed. 2025-26, 11th, Page No. 235)

[Contd...

10. (2)



In series, $k_{eq} = \dfrac{L_1 + L_2}{\dfrac{L_1}{k_a} + \dfrac{L_2}{k_b}}$

$= \dfrac{d+d}{\dfrac{d}{k_a} + \dfrac{d}{k_b}} = \dfrac{2k_a k_b}{k_a + k_b}$

(NCERT Ed. 2025-26, 11th, Page No. 215)

11. (1)

According to Wein's law, $\lambda_m \propto \dfrac{1}{T}$ when temperature is increased, the peak emission wavelength of a black body shifts to lower wavelength.

(NCERT Ed. 2025-26, 11th, Page No. 218)

12. (3)

$\vec{v}_2 - \vec{v}_1 = \vec{u}_1 - \vec{u}_2$

$-\hat{i} - \vec{v}_1 = 2\hat{i} - (-\hat{i})$

$\vec{v}_1 = -4\hat{i}$ m/s

(NCERT Ed. 2025-26, 11th, Page No. 85)

13. (1)

$T^2 = \dfrac{4\pi^2}{GM} r^3$

So slope is $\dfrac{4\pi^2}{GM}$

(NCERT Ed. 2025-26, 11th, Page No. 137)

14. (4)

$W = \Delta k$

$kt = \dfrac{1}{2}mv^2 - 0$

$\Rightarrow v = \sqrt{\dfrac{2kt}{m}}$

$a = \dfrac{dv}{dt} = \sqrt{\dfrac{k}{2mt}}$

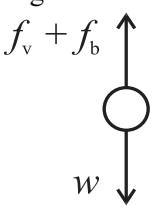
$F = ma$

$= \sqrt{\dfrac{km}{2t}}$

(NCERT Ed. 2025-26, 11th, Page No. 83)

15. (3)

Fbd of body during motion in fluid is:



Where w is weight equal to mg and f_v is viscous force equal to $6\pi\eta rv$ and f_b is buoyant force.

(NCERT Ed. 2025-26, 11th, Page No. 192)

16. (2)

Here final and initial positions are same.

$W_g = 0$

$\Delta k = W_{all} = W_g + W_{air}$

$\dfrac{1}{2}m\left(v^2 - u^2\right) = 0 + W_{air}$

$\Rightarrow W_{air} = -32$J

(NCERT Ed. 2025-26, 11th, Page No. 76)

17. (4)

B: 530 Hz



A: 536 Hz 524 Hz

Frequency $\propto \sqrt{\text{Tension}}$

If tension in B is slightly decreased, then frequency of B decreases and beats increases therefore frequency of A will be 536 Hz.

(NCERT Ed. 2025-26, 11th, Page No. 294)

18. (2)

$B \rightarrow A$

$\Delta Q = 0$

$0 = -30 + \Delta U_{BA}$

$\Delta U_{BA} = 30$ J

$\therefore \Delta U_{AB} = -\Delta U_{BA} = -30$ J

(NCERT Ed. 2025-26, 11th, Page No. 230)

19. (2)

work done at constant pressure is

$W = p\Delta V = nR\Delta T$

Given, $n = 6$, $T_2 = 80°C = 353K$

$T_1 = 30°C = 303$ K, $R = 8.3$ J/mol-K

$\therefore \quad W = 6 \times 8.3 \times (50) \Rightarrow W = 2.5$ KJ

(NCERT Ed. 2025-26, 11th, Page No. 235)

20. (3)

Pitch = 0.5 mm, $LC = \dfrac{0.5 \text{ mm}}{100} = 0.005$ mm

(1) ZE : $\Rightarrow$ +ve

ZE = + 2 div × 0.005 mm = 0.010 mm

(2) Measurement:

MSR = 0.5 mm × 8 = 4.0 mm

CSR = 83 × 0.005 mm = 0.415 mm

Corrected reading

= 4.0 mm + 0.415 mm – (+ 0.010) mm

= 4.405 mm.

(NCERT Ed. 2025-26, 11th, Page No. 11)

[Contd...

21. (2)

When body going vertically upwards, reaches at the highest point, then it is momentarily at rest and it then reverses its direction. At the highest point of motion, its velocity is zero but its acceleration is equal to acceleration due to gravity.

(NCERT Ed. 2025-26, 11th, Page No. 23)

22. (2)

$\rho_{oil} h_{oil} = \rho_{water} h_{water}$

$\Rightarrow \rho_{oil} = \frac{1000(6)}{8} = 750 \text{ kg m}^{-3}$

(NCERT Ed. 2025-26, 11th, Page No. 184)

23. (3)

$v^2 = 0^2 + 2(\alpha r)(n \times 2\pi r)$

$\Rightarrow \alpha = \frac{v^2}{4\pi n r^2} = \frac{3}{\pi r^2} \ [n = 3]$

(NCERT Ed. 2025-26, 11th, Page No. 117)

24. (1)

$\frac{P}{\rho} = \frac{RT}{M_W} \Rightarrow \rho = \frac{PM_W}{RT}$

$\rho = \frac{498 \times 10^3 \times 2 \times 10^{-3}}{8.3 \times 400}$

$\rho = 0.3 \text{ kg/m}^3$

(NCERT Ed. 2025-26, 11th, Page No. 246)

25. (2)

Work done

$= \frac{1}{2} \times \text{stress} \times \text{strain} \times \text{volume}$

$= \frac{1}{2} \times Y \times (\text{strain})^2 \times Al$

$= \frac{1}{2} \times Y \times \frac{x^2}{l^2} \times Al$

$= \frac{Yx^2 A}{2l}$

(NCERT Ed. 2025-26, 11th, Page No. 174)

26. (1)

Escape velocity is independent of the launch angle.

(NCERT Ed. 2025-26, 11th, Page No. 136)

27. (2)

To cross the river exactly opposite starting point, component of velocity of swimmer w.r.t. ground should be zero along stream line.

$4\cos\theta + 2 = 0$

$\cos\theta = -\frac{1}{2}$

(NCERT Ed. 2025-26, 11th, Page No. 34)

28. (2)

$f = \frac{n}{2l}\sqrt{\frac{T}{\mu}} = \frac{5}{2 \times 1.5}\sqrt{\frac{36}{1}} = 5 \times 2 = 10 \text{Hz}$

(NCERT Ed. 2025-26, 11th, Page No. 291)

29. (1)

$K = \sqrt{\frac{I}{M}} = \sqrt{\frac{Ml^2/12}{M}} = \frac{l}{\sqrt{12}}$

(NCERT Ed. 2025-26, 11th, Page No. 116)

30. (3)

$Q_1 = 10 \times 1 \times 10 = 100 \text{ cal}$

$Q_2 = 10 \times 0.50[0 - (-20)] + 10 \times 80$

$= (100 + 800)\text{cal} = 900\text{cal}$

As $Q_1 < Q_2$, so ice will not completely melt and final temperature = 0°C

As heat given by water in cooling up to 0°C is only just sufficient to increase the temperature from –20°C to 0°C, hence mixture in equilibrium will consist of 10gm of ice and 10gm of water.

(NCERT Ed. 2025-26, 11th, Page No. 210)

31. (3)

$PV = \mu RT$ or $PV = \frac{m}{M}RT$

For the given data, mT = constant

$4(273 + 12) = m(273 + 50)$

Or $m = \frac{4 \times 285}{323} = 3.53 \text{ g}$

Mass of gas escaped = (4 – 3.53)g = 0.47 g

(NCERT Ed. 2025-26, 11th, Page No. 246)

32. (2)

If the lift has a downward acceleration, then the person will find his weight less than actual weight, because normal force with the floor of the lift will decrease.

(NCERT Ed. 2025-26, 11th, Page No. 69)

[Contd...

33. (2)

$y = A_0 + A\sin\omega t + B\cos\omega t$

Equation of SHM $y' = y - A_0 = A\sin\omega t + B\cos\omega t$

Resultant amplitude,

$A' = \sqrt{A^2 + B^2}$

(NCERT Ed. 2025-26, 11th, Page No. 262)

34. (2)

$\vec{L} = \vec{r} \times \vec{P} \Rightarrow \vec{L} = mvb(-\hat{k})$

(NCERT Ed. 2025-26, 11th, Page No. 109)

35. (3)

$\left|\vec{E}_g\right| = G(2)\left(\frac{1}{2^2} + \frac{1}{4^2} + \frac{1}{8^2} + ...\infty\right)$

$E_g = 2G\left(\frac{\frac{1}{4}}{1-\frac{1}{4}}\right) = \frac{2G}{3}$

(NCERT Ed. 2025-26, 11th, Page No. 130)

36. (3)

$F\cos\theta = \mu(mg - F\sin\theta)$

$F(\cos\theta + \mu\sin\theta) = \mu mg$

$\frac{dF}{d\theta} = \mu mg\left[\frac{-1(-\sin\theta + \mu\cos\theta)}{(\cos\theta + \mu\sin\theta)^2}\right]$

$\frac{dF}{d\theta} = 0 \Rightarrow \sin\theta - \mu\cos\theta = 0$

$\Rightarrow \tan\theta = \mu \Rightarrow \cos\theta = \frac{1}{\sqrt{1+\mu^2}}$

$F_{\min} = \frac{\mu mg}{\cos\theta + (\tan\theta)\sin\theta} = \mu mg\left(\frac{1}{\sqrt{1+\mu^2}}\right)$

(NCERT Ed. 2025-26, 11th, Page No. 60)

37. (3)

$k_{eff} = k + k + k = 3k$

$T = 2\pi\sqrt{\frac{m}{k_{eff}}} = 2\pi\sqrt{\frac{m}{3k}}$

(NCERT Ed. 2025-26, 11th, Page No. 268)

38. (2)

$v = \frac{\omega}{k} = \frac{18}{4.5} = 4\text{ m/s}$

$4 = \sqrt{\frac{T}{1.3\times10^{-4}}}$

$T = 16\times1.3\times10^{-4} = 20.8\times10^{-4}\text{ N}$

(NCERT Ed. 2025-26, 11th, Page No. 285)

39. (2)

$v_{rms} \propto \sqrt{\frac{T}{M}}$

$\frac{1930}{x} = \sqrt{\frac{300}{1200}\times\frac{32}{2}}$

$x = \frac{1930}{2} = 965\text{ m/s}$

(NCERT Ed. 2025-26, 11th, Page No. 250)

40. (2)

Exponents are insignificant. So, number of significant figures are 3.

(NCERT Ed. 2025-26, 11th, Page No. 4)

41. (1)

Hooke's law is valid till proportionality limit. Bulk modulus is defined for all the three states of matter. In a rod having mass, the normal stress changes at every cross-section.

(NCERT Ed. 2025-26, 11th, Page No. 169)

42. (2)

$v_A = \omega R,\ v_B = 2\omega R$

$v_{B/A} = v_A + v_B = 3\omega R$

(NCERT Ed. 2025-26, 11th, Page No. 41)

43. (1)

A is correct but R is false.

(NCERT Ed. 2025-26, 11th, Page No. 191)

44. (1)

A-III: First Law defines the geometry of the path (Law of Orbits).

C-II: Third Law relates time and distance: $T^2 \propto a^3$.

B-I: Areal velocity is mathematically defined as $\frac{dA}{dt} = \frac{L}{2m}$, related to kepler's second law.

(NCERT Ed. 2025-26, 11th, Page No. 128)

45. (2)

$u_{\text{balloon}} = 0\text{ m/s}$

$v = u + at$

$v = 0 + (4\times10)$

$v = 40$ m/s

Initial velocity of stone = 40 m/s

$v_{\text{stone}} = u_{\text{stone}} - gt$

$= 40 - (10 \times 3) = 10$ m/s

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[Contd...

46. (3)

- 22.4 L of CO_2 weigh at STP = 44 g
  11.2 L will weigh = 22 g
- 22.4 L of N_2 weigh at STP = 28 g
  11.2L will weigh = 14 g
- 22.4 L of NO_2 weigh at STP = 46 g
  11.2L will weigh = 23 g
- 22.4 L of CO weigh at STP = 28 g
  11.2 L will weigh = 14 g

(NCERT Ed. 2025-26, 11th Part-I Page No. 20)

47. (4)

For H-atom:

$E_2 - E_1 > E_3 - E_2 > E_4 - E_3 > E_5 - E_4$

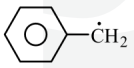
(NCERT Ed. 2025-26, 11th Part-I Page No. 48)

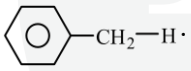
48. (3)

Z = 51(Sb), belongs to group-15 and period-5 of modern periodic table.

(NCERT Ed. 2025-26 11th Part-I Page No. 79)

49. (1)

$CH_2=CH-\dot{C}H_2$ is more stable than 

because C–H bond dissociation energy in $CH_2=CH-CH_2-H$ is less than C–H bond dissociation energy in 

(NCERT Ed. 2025-26, 11th, Part-II Page No. 272)

50. (1)

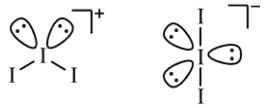
Due to inert pair effect, 13th group element shows +1 oxidation state.

51. (4)

D → E represents isobaric process.

(NCERT Ed. 2025-26 11th Part-I Page No. 140)

52. (4)



(sp^3) (sp^3d)

(NCERT Ed. 2025-26 11th Part-I Page No. 114)

53. (4)

In SO_2, S is present in +4 oxidation state so it can be oxidised and reduced as its minimum and maximum oxidation states are –2 and +6 respectively.

(NCERT Ed. 2025-26 11th Part-II Page No. 241)

54. (2)

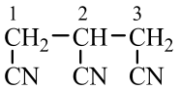
$N_2(g) + 3H_2(g) \rightleftharpoons NH_3(g)$

$\Delta n_g = 2 - 4 = -2$

$K_p = K_c(RT)^{-2}$

(NCERT Ed. 2025-26 11th Part-I Page No. 176)

55. (1)



Propane-1,2,3-tricarbonitrile

(NCERT Ed. 2025-26 11th Part-II Page No. 267)

56. (4)

Position of double bound is different

(NCERT Ed. 2025-26, 11th, Part-II Page No. 270)

57. (1)

Under oxidative workup any terminal alkene $(R-CH=CH_2)$ or terminal alkyne (R–C≡CH) gives CO_2 because the $=CH_2$ or ≡CH carbon is ultimately oxidized fully to CO_2 Internal (non-terminal) C=C or C≡C carbons given carboxylic acids (or ketones), not CO_2.

(NCERT Ed. 2025-26, 11th, Page No. 313)

58. (2)

Law of conservation of mass is invalid for nuclear fission as it involves conversion of mass into energy.

(NCERT Ed. 2025-26, 11th Part-I Page No. 14)

59. (1)

The statement the electron velocity and position in the orbit cannot be determined simultaneously does not belong to Bohr's model of hydrogen atom.

Photoelectric effect is the emission of electrons from a metal surface when exposed to light or electromagnetic radiation of a sufficiently high frequency

(NCERT Ed. 2025-26, 11th Part-I Page No. 46)

60. (2)

The chemistry of early actinoids is more complicated than the corresponding lanthanoids. It is due to large number of oxidation states possible for these actinoids elements.

(NCERT Ed. 2025-26 11th Part-I Page No. 84)

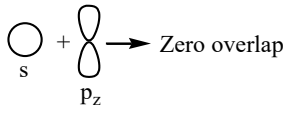
61. (2)

Isothermal	$q = -w$
Isobaric	$q = nC_p\Delta T$
Adiabatic	$q = 0$
Isochoric	$q = \Delta U$

(NCERT Ed. 2025-26 11th Part-I Page No. 140)

[Contd...

62. (1)



(NCERT Ed. 2025-26 11th Part-I Page No. 119)

63. (1)

$$6Fe^{2+} + Cr_2O_7^{2-} + 14H^+ \rightarrow 6Fe^{3+} + 2Cr^{3+} + 7H_2O$$

$$x - y + z = 6 - 1 + 14 = 19$$

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64. (4)

- If K_c is very large ($> 10^3$), the reversible reaction proceeds nearly to completion.
- If K_c is very small ($< 10^{-3}$), the reversible reaction proceeds rarely.

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65. (2)

An alkyl cyanide and its corresponding alkyl isocyanide both have the same number of each type of atom but differ in which atom of the –C≡N unit is attached to the alkyl group. Thus they possess different functional groups, making them functional group isomers, and the reason accurately explains that this difference in connectivity of carbon and nitrogen leads to the isomerism.

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66. (4)

- Molarity of pure water:
  1 L of water (ρ = 1 g/mL) weighs 1000 g.

  $$\text{Moles of } H_2O = \frac{1000}{18} = 55.55$$

  $$M = \frac{n}{v(L)} = \frac{55.55}{1} = 55.55\ M$$

- Molarity of 1 mole of solute in 1 L of solution:

  $$M = \frac{n}{v(L)} = \frac{1}{1} = 1\ M$$

- Molarity of 49 g of H_2SO_4 dissolved in 100 mL of solution:

  $$M = \frac{n}{v(mL)} \times 1000 = \frac{49 \times 1000}{98 \times 100} = 5\ M$$

- Molarity of 1 mole of NaOH dissolved in 100 g of solution (ρ of solution = 1.2 g/mL):

  $$\text{Volume of solution} = \frac{100}{1.2}$$

  $$M = \frac{n}{v(mL)} \times 1000 = \frac{1 \times 1.2}{100} \times 1000 = 12\ M$$

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67. (4)

If KE = K, there PE = –2 K

$$\therefore \frac{PE}{KE} = -\frac{2K}{K} = -2\ ;\quad \frac{KE}{PE} = -\frac{1}{2}$$

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68. (1)

First trans-uranic element (element after uranium) is Np (Z = 93).

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69. (1)

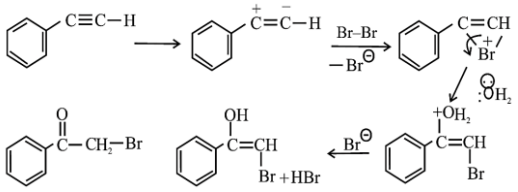
Formal charge =

$$\left[\begin{matrix}\text{Total number of valence}\\ \text{electrons in the free atom}\end{matrix}\right] - \left[\begin{matrix}\text{Total number of lone pair}\\ \text{electrons}\end{matrix}\right] - \frac{\left[\begin{matrix}\text{Total number of}\\ \text{bonding electrons}\end{matrix}\right]}{2}$$

- Formal charge on oxygen 1: $6 - 2 - \frac{6}{2} = +1$
- Formal charge on oxygen 2: $6 - 4 - \frac{4}{2} = 0$
- Formal charge on oxygen 3: $6 - 6 - \frac{2}{2} = -1$

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70. (3)



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71. (3)

- 0.00250: Three significant figures
- 0.63000: Five significant figures
- 7.0004: Five significant figures
- 1.0000: Five significant figures

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72. (3)

$$\text{Wave number } (\bar{\nu}) = \frac{1}{\lambda}$$

so, unit of $\bar{\nu}$ is m^{-1}.

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73. (1)

I, II, IV are correct by definition. III is false: naphthalene contains fused benzene rings → benzenoid.

(NCERT Ed. 2025-26, 11th Part-II Page No. 321)

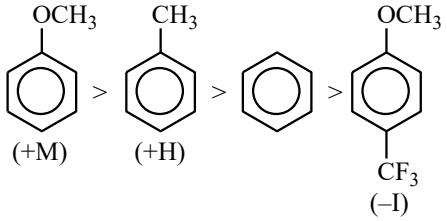
74. (3)

- Boron (B): 88 pm (metallic radius)
- Aluminium (Al): 143 pm
- Gallium (Ga): 135 pm
- Indium (In): 167 pm

[Contd...

75. (2)

Higher the electron density on benzene ring, higher its reactivity towards electrophilic substitution reaction. Donation by +M effect is more dominating than donation by +11 and +1 effect.



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76. (2)

Kjeldahl method cannot estimate nitrogen in nitro group, Azo group and N in ring.

(NCERT Ed. 2025-26, 11th Part-II Page No. 287)

77. (2)

IV is false: Pb(IV) is less stable than Pb(II), hence it acts as an oxidising agent by getting reduced to Pb(II).

78. (1)

I–III capture the definition and implication of state variables. IV is false (it describes a path function, not a state function).

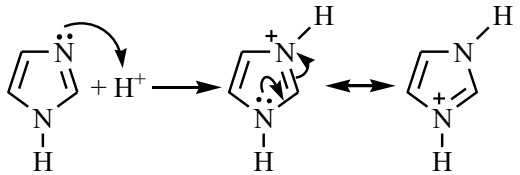
(NCERT Ed. 2025-26, 11th Part-I Page No. 138)

79. (3)

B is least basic as lone pair of electron is involved in resonance.

D is the most basic because it results in formation of equivalent resonating structures when H- attack on it.

Between A and C, the former is less basic as hybridisation of nitrogen is sp^2 which decreases its basic strength.



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80. (1)

As PV = nRT, where n and T are constant

$PV_m = k = nRT$ … (i)

isothermal expansion $PV_m = k$ (graph → C)

$P = \frac{k}{V_m} \rightarrow$ (Graph A)

(NCERT Ed. 2025-26, 11th Part-I Page No. 141)

81. (2)

In paper chromatography, water held in the pores/cellulose acts as the stationary phase (partition chromatography).

(NCERT Ed. 2025-26, 11th Part-II Page No. 283)

82. (2)

In acidified aqueous medium Al^{3+} exists mainly as $[Al(H_2O)_6]^{3+}$, which is octahedral.

83. (2)

Oxidation state of Cr remains +6 in chromate and dichromate (interconversion is acid–base equilibrium, not redox).

(NCERT Ed. 2025-26, 11th Part-II Page No. 246)

84. (1)

No common ions present, so AgCl has maximum solubility (common ion effect lowers solubility).

(NCERT Ed. 2025-26, 11th Part-I Page No. 201)

85. (2)

trigonal planar, trigonal planar, linear – Electron-pair geometry: SO_2 and NO_2^- are trigonal planar (e^- geometry); N_3^- is linear.

(NCERT Ed. 2025-26, 11th Part-I Page No. 112)

86. (3)

These are not valid reasons: (C) is incorrect as stated, and (E) is not taken as a primary cause of half-filled stability.

(NCERT Ed. 2025-26, 11th Part-I Page No. 65)

87. (3)

Same logic: equal K_a and K_b ⇒ hydrolysis effects cancel ⇒ neutral.

(NCERT Ed. 2025-26, 11th Part-I Page No. 201)

88. (3)

Paramagnetic species among given set are those with unpaired π* electrons (e.g., O_2, O_2^+, O_2^-, N_2^{2-}, N_2^+, N_2^- types), giving total 6.

(NCERT Ed. 2025-26, 11th Part-I Page No. 128)

89. (4)

Same reason: maximum heat comes from largest moles of strong–strong neutralisation among options.

(NCERT Ed. 2025-26, 11th Part-I Page No. 156)

90. (1)

B and D are d block elements

(NCERT Ed. 2025-26, 11th Part-I Page No. 84)

[Contd...

91. (3)

Systematic and monumental description of life forms brought in, out of necessity, detailed systems of identification, nomenclature and classification. The biggest spin off of such studies was the recognition of the sharing of similarities among living organisms both horizontally and vertically. That all present day living organisms are related to each other and also to all organisms that ever lived on this earth, was a revelation which humbled man and led to cultural movements for conservation of biodiversity.

(NCERT Ed. 2022-23, 11th, Page No. 1)

92. (2)

Using a prism T. W. Engelmann split light into its spectral components and then illuminated a green alga, *Cladophora*, placed in a suspension of aerobic bacteria.

(NCERT Ed. 2025-26, 11th, Page No. 135)

93. (1)

The specialised kind of cell division that reduces the chromosome number by half results in the production of haploid daughter cells. This kind of division is called meiosis.

(NCERT Ed. 2025-26, 11th, Page No. 125)

94. (4)

The tissue between the upper and the lower epidermis in a dorsiventral leaf is called the mesophyll. Mesophyll, which possesses chloroplasts and carry out photosynthesis, is made up of parenchyma (tissue with living cells).

(NCERT Ed. 2025-26, 11th, Page No. 76)

95. (1)

Rust and smut in plants are caused by the fungi *Puccinia* and *Ustilago*, respectively. Malaria in humans is caused by a protist, *Plasmodium*. Canker in citrus is a bacterial disease. Mosaic in tobacco is caused by a virus.

(NCERT Ed. 2025-26, 11th, Page No. 14)

96. (3)

The simplest plant growth hormone is ethylene. It is synthesised in large amounts by tissues undergoing senescence and ripening fruits. The most widely used compound as source of ethylene is ethephon. More than 100 types of gibberellins have been isolated from plants and fungi. Auxin was first isolated from human urine.

(NCERT Ed. 2025-26, 11th, Page No. 177)

97. (2)

The plasma membrane is selectively permeable to some molecules present on either side of it. Many molecules can move briefly across the membrane without any requirement of energy and this is called the passive transport. Neutral solutes may move across the membrane by the process of simple diffusion along the concentration gradient, i.e., from higher concentration to the lower. Water may also move across this membrane from higher to lower concentration. Movement of water by diffusion is called osmosis. As the polar molecules cannot pass through the nonpolar lipid bilayer, they require a carrier protein of the membrane to facilitate their transport across the membrane. A few ions or molecules are transported across the membrane against their concentration gradient, i.e., from lower to the higher concentration. Such a transport is an energy dependent process, in which ATP is utilised and is called active transport.

(NCERT Ed. 2025-26, 11th, Page No. 94)

98. (4)

In most higher plants, the growing apical bud inhibits the growth of the lateral (axillary) buds, a phenomenon called apical dominance. Removal of shoot tips (decapitation) usually results in the growth of lateral buds. It is widely applied in tea plantations, hedge-making.

(NCERT Ed. 2025-26, 11th, Page No. 175)

99. (3)

Phaeophyceae - Food is stored as complex carbohydrates, which may be in the form of laminarin or mannitol. 2-8 equal, apical flagella is typically a characteristic of chlorophyceae, rhodophyceae members lack flagella but possess r-phycoerythrin and store food as floridean starch.

(NCERT Ed. 2025-26, 11th, Page No. 27)

100. (1)

When fats are used in respiration, the RQ is less than 1.

(NCERT Ed. 2025-26, 11th, Page No. 164)

101. (3)

Monocotyledonous roots do not undergo any secondary growth.

(NCERT Ed. 2025-26, 11th, Page No. 74)

102. (2)

The correct ascending order of the given taxonomic categories for mango is:

D. *Mangifera* (Genus)
C. Anacardiaceae (Family)
B. Sapindales (Order)
E. Angiospermae (Division)
A. Plantae (Kingdom)

Therefore, the ascending order is D, C, B, E, A.

(NCERT Ed. 2025-26, 11th, Page No. 8)

[Contd...

103. (4)
RuBisCO (Ribulose-1,5-bisphosphate carboxylase /oxygenase) is an enzyme that catalyzes the first major step of carbon fixation in the Calvin cycle. In this step, it catalyzes the reaction between ribulose-1,5-bisphosphate (RuBP) and carbon dioxide (CO_2) to form two molecules of 3-phosphoglycerate. This addition of a carboxyl group (from CO_2) is a carboxylation reaction. While RuBisCO also has oxygenase activity (leading to photorespiration), its primary and intended role in the Calvin cycle for carbon fixation is that of a carboxylase.
(NCERT Ed. 2025-26, 11th, Page No. 143)

104. (1)
During pachytene, the four chromatids of each bivalent chromosomes becomes distinct and clearly appears as tetrads. This stage is characterised by the appearance of recombination nodules, the sites at which crossing over occurs between non-sister chromatids of the homologous chromosomes. Recombination between homologous chromosomes is completed by the end of pachytene, leaving the chromosomes linked at the sites of crossing over. The first two stages of prophase I are relatively short-lived compared to the next stage that is pachytene.
(NCERT Ed. 2025-26, 11th, Page No. 126)

105. (2)
When the placenta is axial and the ovules are attached to it in a multilocular ovary, the placentation is said to be axile, as in china rose, tomato and lemon.
(NCERT Ed. 2025-26, 11th, Page No. 65)

106. (3)
Methanogens are classified as archaebacteria.
(NCERT Ed. 2025-26, 11th, Page No. 13)

107. (4)
The term "amphibians of the plant kingdom" refers to Bryophytes. In the given question, *Polytrichum*, *Marchantia* and *Sphagnum* are bryophytes. *Adiantum* is a pteridophyte and *Cedrus* is a gymnosperm.
(NCERT Ed. 2025-26, 11th, Page No. 29)

108. (1)
The golgi apparatus consists of many flat, disc-shaped sacs or cisternae. These are stacked parallel to each other. The isolated lysosomal vesicles have been found to be very rich in almost all types of hydrolytic enzymes. The vacuole is bound by a single membrane called tonoplast. Ribosomes are the non-membrane bound organelles.
(NCERT Ed. 2025-26, 11th, Page No. 95)

109. (4)
Oxygen acts as the final hydrogen acceptor in the mitochondrial ETS.
(NCERT Ed. 2025-26, 11th, Page No. 160)

110. (3)
In arithmetic growth, following mitotic cell division, only one daughter cell continues to divide while the other differentiates and matures. On plotting the length of the organ against time, a linear curve is obtained. In most systems, the initial growth is slow (lag phase), and it increases rapidly thereafter – at an exponential rate (log or exponential phase). Here, both the progeny cells following mitotic cell division retain the ability to divide and continue to do so. However, with limited nutrient supply, the growth slows down leading to a stationary phase. If we plot the parameter of growth against time, we get a typical sigmoid or S-curve. This is called geometric growth.
(NCERT Ed. 2025-26, 11th, Page No. 170)

111. (2)
In a dicot root, next to endodermis lies a few layers of thick-walled parenchymatous cells referred to as pericycle. Initiation of lateral roots and vascular cambium during the secondary growth takes place in these cells.
(NCERT Ed. 2025-26, 11th, Page No. 74)

112. (1)
In C_4 plants, the primary CO_2 fixation product is a 4-carbon organic acid viz. oxaloacetic acid or OAA. Though these plants have the C_4 oxaloacetic acid as the first CO_2 fixation product they use the C_3 pathway or the Calvin cycle as the main biosynthetic pathway.
(NCERT Ed. 2025-26, 11th, Page No. 145)

113. (3)
It is significant to note that in the 24 hour average duration of cell cycle of a human cell, cell division proper lasts for only about an hour. The interphase (resting phase) lasts more than 95% of the duration of cell cycle.
(NCERT Ed. 2025-26, 11th, Page No. 121)

114. (4)
The two kingdom classification system did not distinguish between the eukaryotes and prokaryotes, unicellular and multicellular organisms and photosynthetic (green algae) and non-photosynthetic (fungi) organisms. Hence the two kingdom classification used for a long time was found inadequate. The three-domain system has also been proposed that divides the Kingdom Monera into two domains, leaving the remaining eukaryotic kingdoms in the third domain and thereby a six kingdom classification.
(NCERT Ed. 2025-26, 11th, Page No. 10)

[Contd...

115. (4)

Genus comprises a group of related species which has more characters in common in comparison to species of other genera.

(NCERT Ed. 2025-26, 11th, Page No. 7)

116. (3)

In some plants such as *Rhizophora* growing in swampy areas, many roots come out of the ground and grow vertically upwards. Such roots, called pneumatophores, help to get oxygen for respiration.

(NCERT Ed. 2025-26, 11th, Page No. 58)

117. (2)

Unlike bryophytes and pteridophytes, in gymnosperms the male and the female gametophytes do not have an independent free-living existence. The multicellular female gametophyte is also retained within megasporangium.

(NCERT Ed. 2025-26, 11th, Page No. 33)

118. (1)

The nuclear envelope, which consists of two parallel membranes with a space between (10 to 50 nm) called the perinuclear space, forms a barrier between the materials present inside the nucleus and that of the cytoplasm. The outer membrane usually remains continuous with the endoplasmic reticulum and also bears ribosomes on it. These nuclear pores are the passages through which movement of RNA and protein molecules takes place in both directions between the nucleus and the cytoplasm. A single human cell has approximately two metre long thread of DNA distributed among its forty six (twenty three pairs) chromosomes. The content of nucleolus is continuous with the rest of the nucleoplasm as it is not a membrane bound structure. It is a site for active ribosomal RNA synthesis.

(NCERT Ed. 2025-26, 11th, Page No. 100)

119. (4)

Hexokinase: This enzyme catalyzes the first step of glycolysis, converting glucose to glucose-6-phosphate. Glycolysis is the initial pathway for glucose breakdown in both aerobic and anaerobic respiration. Therefore, hexokinase is involved.

In fermentation, say by yeast, the incomplete oxidation of glucose is achieved under anaerobic conditions by sets of reactions where pyruvic acid is converted to CO_2 and ethanol. The enzymes, pyruvic acid decarboxylase and alcohol dehydrogenase catalyse these reactions.

Citrate synthase: This enzyme catalyzes the first step of the Krebs' cycle (citric acid cycle), combining acetyl-CoA with oxaloacetate to form citrate. The Krebs' cycle is a major part of aerobic respiration. Under anaerobic conditions, yeast does not convert pyruvate to acetyl-CoA and does not perform the Krebs' cycle. Therefore, citrate synthase is not involved.

(NCERT Ed. 2025-26, 11th, Page No. 156)

120. (2)

All tissues except epidermis and vascular bundles constitute the ground or fundamental tissue system. It consists of simple tissues such as parenchyma, collenchyma and sclerenchyma. The epidermal tissue system bears trichomes and root hairs.

(NCERT Ed. 2025-26, 11th, Page No. 72)

121. (4)

Abscisic acid is a derivative of carotenoids.

(NCERT Ed. 2025-26, 11th, Page No. 174)

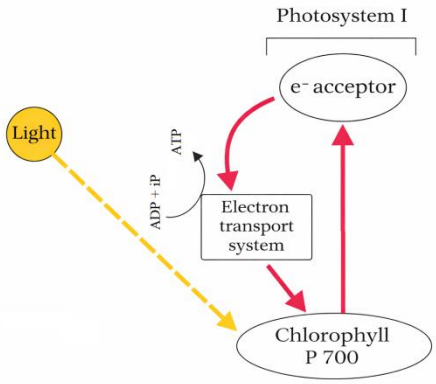
122. (1)

Katherine Esau reported that the curly top virus spreads through the phloem.

(NCERT Ed. 2025-26, 11th, Page No. 56)

123. (3)

When only PS I is functional, the electron is circulated within the photosystem and the phosphorylation occurs due to cyclic flow of electrons. A possible location where this could be happening is in the stroma lamellae. Cyclic photophosphorylation also occurs when only light of wavelengths beyond 680 nm are available for excitation. In PS I the reaction centre chlorophyll *a* has an absorption peak at 700 nm, hence is called P700, while in PS II it has absorption maxima at 680 nm, and is called P680. The electrons needed to replace those removed from photosystem I are provided by photosystem II.



Cyclic photophosphorylation

(NCERT Ed. 2025-26, 11th, Page No. 140)

[Contd...

124. (4)
With the exception of yeasts which are unicellular, fungi are filamentous. They prefer to grow in warm and humid places. Fusion of protoplasms between two motile or non-motile gametes called plasmogamy. Fusion of two nuclei called karyogamy.
(NCERT Ed. 2025-26, 11th, Page No. 16)

125. (1)
When stamens are attached to the petals, they are epipetalous as in brinjal, or epiphyllous when attached to the perianth as in the flowers of lily. The stamens may be united into one bunch or one bundle (monoadelphous) as in china rose, or two bundles (diadelphous) as in pea, or into more than two bundles (polyadelphous) as in citrus.
(NCERT Ed. 2025-26, 11th, Page No. 64)

126. (2)
Prophase is marked by the initiation of condensation of chromosomal material. The key features of metaphase are: Chromosomes are moved to spindle equator and get aligned along metaphase plate. Anaphase stage is characterised by the following key events: Centromeres split and chromatids separate. In telophase, chromosomes cluster at opposite spindle poles and their identity is lost as discrete elements.
(NCERT Ed. 2025-26, 11th, Page No. 122)

127. (3)
The meristem which occurs between mature tissues is known as intercalary meristem. They occur in grasses and regenerate parts removed by the grazing herbivores. In grasses, the guard cells are dumb-bell shaped. In grasses, certain adaxial epidermal cells along the veins modify themselves into large, empty, colourless cells. These are called bulliform cells. The mesophyll of dorsiventral leaf has two types of cells – the palisade parenchyma and the spongy parenchyma. Grass leaf is isobilateral.
(NCERT Ed. 2025-26, 11th, Page No. 77)

128. (2)
Pteridophyte genera like *Selaginella* and *Salvinia* which produce two kinds of spores, macro (large) and micro (small) spores, are known as heterosporous. The megaspores and microspores germinate and give rise to female and male gametophytes, respectively. The female gametophytes in these plants are retained on the parent sporophytes for variable periods. The development of the zygotes into young embryos take place within the female gametophytes. This event is a precursor to the seed habit considered an important step in evolution.
(NCERT Ed. 2025-26, 11th, Page No. 32)

129. (4)
ER helps in the transport of substances, synthesis of proteins, lipoproteins and glycogen.
(NCERT Ed. 2025-26, 11th, Page No. 95)

130. (1)

Process	Product
Glycolysis	Pyruvic acid
Calvin cycle	Sugar
Photorespiration	Phosphoglycolate and phosphoglycerate
Fermentation in animal muscle cells	Lactic acid

(NCERT Ed. 2025-26, 11th, Page No. 147)

131. (3)
Charles Darwin and his son Francis Darwin observed that the coleoptiles of canary grass responded to unilateral illumination by growing towards the light source (phototropism).
(NCERT Ed. 2025-26, 11th, Page No. 174)

132. (4)
Viroids are free RNA; lack the protein coat that is found in viruses.
(NCERT Ed. 2025-26, 11th, Page No. 21)

133. (2)
Poaceae is a monocot family. The members produce one seeded fruit called caryopsis (grain).
(NCERT Ed. 2025-26, 11th, Page No. 66)

134. (4)
A chromatographic separation of the leaf pigments shows that the colour that we see in leaves is not due to a single pigment but due to four pigments: Chlorophyll *a* (bright or blue green in the chromatogram), chlorophyll *b* (yellow green), xanthophylls (yellow) and carotenoids (yellow to yellow-orange).
(NCERT Ed. 2025-26, 11th, Page No. 137)

135. (1)
The algae reproduce by vegetative, asexual and sexual methods. Vegetative reproduction is by fragmentation. Each fragment develops into a thallus. Asexual reproduction is by the production of different types of spores, the most common being the zoospores. They are flagellated (motile) and on germination gives rise to new plants. Sexual reproduction takes place through fusion of two gametes.
(NCERT Ed. 2025-26, 11th, Page No. 24)

[Contd...

136. (3)

The junction between two neurons is called a synapse. This includes the membrane of a pre-synaptic neuron and a post-synaptic neuron, which may or may not be separated by a physical space. This specific space or gap is called the synaptic cleft.

(NCERT Ed. 2025-26, 11th, Page No. 234)

137. (4)

Amino acids are classified based on the nature of their R-group. Lysine is a basic amino acid because it contains an additional amino group in its side chain, giving it two amino groups in total. Glutamic acid, is an acidic amino acid. Serine, is a neutral amino acid with a hydroxyl group , and Valine, is a standard neutral amino acid.

(NCERT Ed. 2025-26, 11th, Page No. 106)

138. (3)

List-I (Animal)	List-II (Characteristic feature)
Pila	Radula
Bombyx	Malpighian tubules
Pleurobrachia	Comb plates
Asterias	Water vascular system

(NCERT Ed. 2025-26, 11th, Page No. 44)

139. (4)

In male cockroaches, the mushroom gland is a large, characteristic accessory reproductive gland located in the 6th-7th abdominal segments.

Tegmina are the opaque, leathery forewings, not hindwings. Hepatic caeca are found at the junction of the foregut and midgut. Anal styles are present only in males, helping to distinguish them from females.

(NCERT Ed. 2020-21, 11th, Page No. 114)

140. (2)

The description provided—a marine invertebrate with a soft, unsegmented body, a calcareous shell, sensory tentacles, and a radula (the file-like rasping organ)—is a classic definition of the Phylum Mollusca.

(NCERT Ed. 2025-26, 11th, Page No. 44)

141. (2)

The erythrocyte count is actually 5 to 5.5 million per mm^3, not billion. While neutrophils are phagocytic, lymphocytes are responsible for immune responses (producing antibodies or T-cell responses) rather than direct phagocytosis; monocytes are the other primary phagocytic cells. Platelets (thrombocytes) are cell fragments produced from megakaryocytes, not from "thrombocytes" themselves.

(NCERT Ed. 2025-26, 11th, Page No. 194)

142. (2)

While ANF (Atrial Natriuretic Factor) does check the renin-angiotensin mechanism, it does so by causing vasodilation (widening of blood vessels) to decrease blood pressure, not vasoconstriction.

The concentration of urine depends on the counter-current mechanism involving the Loop of Henle and vasa recta.

(NCERT Ed. 2025-26, 11th, Page No. 212)

143. (2)

When an enzyme acts on a substrate, the substrate must be converted into a transition state structure before the final product is formed. This state is very unstable and transient (short-lived) because it has high internal energy.

(NCERT Ed. 2025-26, 11th, Page No. 115)

144. (2)

Emphysema is a chronic respiratory disorder often caused by cigarette smoking. In this condition, the alveolar walls are damaged, which significantly reduces the surface area available for gas exchange. Asthma involves inflammation of bronchi. Silicosis and Fibrosis are occupational lung diseases caused by inhaling dust or stone particles.

(NCERT Ed. 2025-26, 11th, Page No. 190)

145. (4)

Gastric Inhibitory Peptide (GIP), as the name suggests, inhibits gastric secretion and motility to slow down the process when food reaches the intestine. It does not stimulate them.

(NCERT Ed. 2025-26, 11th, Page No. 247)

146. (3)

While most of the skeleton in vertebrate embryos is made of cartilage, not all of it is replaced by bone in adults. Some cartilage persists in areas like the tip of the nose, outer ear joints, and between adjacent vertebrae. Smooth muscles are fusiform and involuntary, neuroglia provide the bulk of neural volume, and dense regular tissue has parallel collagen fibers.

(NCERT Ed. 2020-21, 11th, Page No. 104)

[Contd...

147. (3)

The ascending limb of Henle's loop is unique because it is impermeable to water but allows the transport of electrolytes (like NaCl) either actively or passively into the medullary fluid.

Reabsorption is actually maximum in the PCT. The descending limb is permeable to water but almost impermeable to electrolytes. Conditional reabsorption occurs in the Distal Convoluted Tubule (DCT).

(NCERT Ed. 2025-26, 11th, Page No. 209)

148. (4)

List-I	List-II
Tight junctions	Help to stop substances from leaking across a tissue
Adhering junctions	Perform cementing to keep neighboring cells together
Gap junctions	Facilitate rapid transfer of ions and small molecules
Squamous epithelium	Involved in forming a diffusion boundary

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149. (3)

During inspiration, the contraction of the diaphragm pulls it downward (flattening it), which increases the volume of the thoracic cavity in the antero-posterior axis. The diaphragm does not act on the dorso-ventral axis. The contraction of external intercostal muscles lifts the ribs and sternum, increasing volume in the dorso-ventral axis. Relaxation of muscles generally leads to expiration, not inspiration.

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150. (2)

Antedon is correctly known as the Sea lily. For the others: *Ophiura* is the Brittle star , Echinus is the Sea urchin , and *Cucumaria* is the Sea cucumber.

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151. (1)

Enzymes are biological catalysts. Unlike inorganic catalysts, which can function effectively at high temperatures and pressures, enzymes are denatured (lose their shape and function) at high temperatures (usually above 40°C). Competitive inhibitors resemble the substrate, not the enzyme. Co-enzymes are usually transiently (temporarily) bound. Some enzymes, like ribozymes, are made of RNA.

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152. (3)

Starch can hold iodine because it forms secondary helical structures. However, cellulose is a linear polymer that does not form these complex helices. Because it lacks these "pockets" to trap iodine molecules, it cannot hold them and remains colorless.

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153. (3)

Phylum Ctenophora (comb jellies) consists of animals that are exclusively marine, diploblastic (two germ layers), and possess the unique ability of bioluminescence (emitting light).

They reproduce only sexually. Digestion is both extracellular and intracellular. Fertilization is external with indirect development.

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154. (1)

The diagram shows a cross-section of a blood vessel i.e., artery. The walls of internal organs like blood vessels, stomach, and intestine contain smooth muscles. These are non-striated and involuntary. Skeletal muscle and voluntary muscle are attached to bones, while cardiac muscle is found only in the heart.

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155. (3)

Tetany is characterized by rapid spasms or wild contractions in muscles. This condition is caused by low Ca^{2+} levels in the body fluids, which increases neuronal excitability. Option (1) describes Gout. Option (2) describes Osteoporosis. Option (4) describes Muscular Dystrophy.

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156. (2)

Micturition (urination) begins when the bladder fills with urine, and stretch receptors on the bladder wall send signals to the CNS. In response, the CNS sends motor signals that cause the smooth muscles of the bladder to contract and simultaneously cause the relaxation of the urethral sphincter, allowing urine to flow out.

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157. (4)

The medulla oblongata is a vital part of the brain stem that controls several involuntary activities. It contains the respiratory center for breathing, the cardiovascular center for heart rate and reflexes, and the center for gastric secretions (along with vomiting and swallowing).

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[Contd...

158. (3)

List-I	List-II
Labrum	Upper lip
Labium	Lower lip
Hypopharynx	Median flexible lobe acting as a tongue
Mandibles	Possess grinding and incising regions

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159. (2)

During muscle contraction, the thin (actin) filaments slide over the thick (myosin) filaments. In a state of maximal contraction, the 'H' zone disappears because the thin filaments from both sides of the sarcomere meet or overlap in the center.

'Z' lines move closer together, the 'A' band remains constant in length and Ca^{2+} is released into the sarcoplasm to trigger contraction, rather than staying in the reticulum.

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160. (1)

List-I	List-II
Uremia	Accumulation of urea in blood due to kidney malfunction
Renal calculi	Mass of crystallized salts (oxalates, etc.) formed within the kidney
Glomerulonephritis	Inflammation of glomeruli of kidney
Hemodialysis	Process of removing urea from blood using an artificial kidney

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161. (4)

The lining of the stomach and intestines is composed of columnar epithelium, which helps in secretion and absorption. PCT (1) has simple cuboidal epithelium with microvilli ; walls of blood vessels (2) have squamous epithelium (endothelium); and bronchioles (3) have ciliated epithelium.

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162. (3)

Lecithin is a phospholipid. Even though it has a low molecular weight (~750 Da), it is found in the acid-insoluble fraction because it forms vesicles/membranes that do not dissolve during tissue grinding. It contains a phosphate group and is a major component of cell membranes.

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163. (4)

List-I	List-II
Fibrinogen	Blood clotting or coagulation
Globulins	Defense mechanisms of the body
Albumins	Osmotic balance
Serum	Plasma without clotting factors

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164. (1)

The image shows a reaction where groups (X and Y) are removed from a substrate without the addition of water, resulting in the formation of a double bond (C=C). This specific type of reaction is catalyzed by Lyases. Transferases move groups between molecules, Hydrolases use water to break bonds, and Ligases link two molecules together.

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165. (4)

Betta is commonly known as the Fighting fish. *Exocoetus* is the Flying fish. *Clarias* is the Magur (catfish). *Pterophyllum* is the Angel fish.

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166. (2)

Growth Hormone (GH), also known as somatotropin, is synthesized and secreted by the Adenohypophysis (the anterior lobe of the pituitary gland). Specifically, it is produced by specialized cells called somatotrophs.

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167. (3)

All vertebrates are chordates (because they possess a notochord during the embryonic stage), but not all chordates are vertebrates (as protochordates like Urochordata and Cephalochordata do not develop a vertebral column). Notochord is throughout life in Cephalochordata and only in larval tail in Urochordata. In chordates, the CNS is dorsal, hollow, and single.

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[Contd...

168. (2)

In male frogs, copulatory pads are found on the first digits of the forelimbs, not the hindlimbs. The other statements are correct: frogs absorb water through the skin, they have a 3-chambered heart, and they are ureotelic.

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169. (1)

Glycogen is a polysaccharide (homopolymer of glucose) used for energy storage in animals. In its branched structure, the right end is called the reducing end, while the left end is called the non-reducing end. It is not a heteropolymer of fructose.

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170. (4)

Each limb in a human (arm or leg) contains exactly 30 bones. Therefore, "both limbs" (referring to a pair, such as both legs or both arms) contain $30 \times 2 = 60$ bones.

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171. (4)

Frogs undergo aestivation (summer sleep) during peak summers and hibernation (winter sleep) during peak winters.

Frogs are poikilotherms (cold-blooded), meaning their body temperature changes with the environment; they cannot maintain a constant temperature.

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172. (3)

The skull is dicondylic but it articulates with atlas (first vertebra) not axis, ribs 8-10 are vertebrochondral, the pectoral girdle has a clavicle and scapula, and the pelvic girdle has two coxal bones.

While the acromion is a process of the scapula, it articulates with the clavicle, not the humerus (the humerus articulates with the glenoid cavity).

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173. (3)

Heart failure is specifically defined as the state where the heart is not pumping blood effectively enough to meet the metabolic needs of the body. It is often called "congestive heart failure" due to lung congestion. It is different from cardiac arrest (heart stops) or a heart attack (muscle damage due to low blood supply). High blood pressure is hypertension.

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174. (3)

The brain is protected by three layers of meninges. The Pia mater is the innermost layer and is very thin and in direct contact with the brain tissue. The Dura mater is the tough outer layer, and the Arachnoid mater is the middle thin layer. White matter is a type of brain tissue, not a meningeal layer.

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175. (4)

The "conducting part" (from nostrils to terminal bronchioles) is responsible for transporting, filtering, and warming air. However, it is not the site for gas diffusion. The actual diffusion of O_2 and CO_2 occurs only in the alveoli and their ducts, which form the "respiratory or exchange part".

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176. (3)

Thymosins primarily promote the differentiation of T-lymphocytes, which are responsible for cell-mediated immunity. B-lymphocytes are associated with humoral immunity (antibodies). The thymus is essential for the immune system, secretes thymosins, and shrinks or degenerates in old age.

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177. (4)

List-I (Excretory structure)	List-II (Representative organism)
Flame cells	*Fasciola*
Nephridia	*Pheretima*
Malpighian tubules	*Locusta*
Antennal glands	Prawn

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178. (2)

Vital Capacity (VC) is the maximum volume of air a person can breathe in after a forced expiration. It includes ERV + TV + IRV. Inspiratory Capacity is TV + IRV. Total Lung Capacity is the total volume in lungs after forced inspiration. Functional Residual Capacity is the air remaining after a normal expiration.

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179. (3)

Steroid hormones (like estrogen or cortisol) are lipid-soluble and can easily cross the cell membrane. They enter the target cell and bind to intracellular receptors (usually in the nucleus) to form a hormone-receptor complex that directly regulates gene expression.

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180. (2)

Connective tissues are the most abundant because they link and support other tissues (this is the real reason for their distribution). While it is true that all connective tissues except blood secrete fibers like collagen, this fact describes their composition rather than explaining why they are the "most widely distributed".

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